

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$P(t) = 290(1.04)^{(\frac{4}{6})t}$$

The function P models the population, in thousands, of a certain city t years after **2005**. According to the model, the population is predicted to increase by $n\%$ every **18** months. What is the value of n ?

Answer

- A. **0.38**
- B. **1.04**
- C. **4**
- D. **6**

Correct Answer: C

Rationale

Choice C is correct. It's given that the function P models the population of the city t years after **2005**. Since there are **12** months in a year, **18** months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x **18**-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P(\frac{18}{12}x) = 290(1.04)^{(\frac{4}{6})(\frac{18}{12}x)}$, which is equivalent to $P(\frac{18}{12}x) = 290(1.04)^x$. Therefore, for each **18**-month period, the predicted population of the city is **1.04** times, or **104%** of, the previous population. This means that the population is predicted to increase by **4%** every **18** months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is **1.04** times the previous year's predicted population, which is not the same as an increase of **1.04%**.

Choice D is incorrect and may result from conceptual or calculation errors.